Technical Writing

Lecture 5

Conducting Research

Conducting Research

Research is essential in technical and professional work. Engineers, scientists, and managers rely on research to understand problems, evaluate solutions, and make informed decisions.

Research helps replace **opinions with evidence**.

Examples of research use:

- Designing a product
- Improving a system
- Solving a technical problem
- Supporting a recommendation in a report

Why Research Is Important

Research prevents **wasted time and resources**.

It helps professionals:

- Understand the real problem
- Learn what solutions already exist
- Avoid repeating previous mistakes
- Make decisions based on evidence

Example:
Before developing a new mobile application, developers study existing apps to understand what works well and what users dislike.

What Is Research?

Research is a **systematic process of collecting and analyzing information to answer a question or solve a problem.**

Key idea:

- Research is organized and planned.
- It uses evidence rather than personal opinion.

Example:
Instead of saying *“this algorithm is faster”*, researchers measure execution time and compare performance.

Research Methods

Research methods are the techniques used to collect and analyze information.

Common research methods include:

- Experiments → testing solutions under controlled conditions
- Surveys → collecting opinions from many people
- Interviews → detailed discussions with participants
- Observations → studying behavior or processes

The stronger the research method, the more reliable the conclusions.

What Is Data?

Data refers to the information collected during research.

Data can include:

- Measurements (temperature, speed, accuracy)
- Observations (how users interact with software)
- Survey responses
- Documents or records

Example:
If researchers test a new battery, they collect data such as charging time, energy capacity, and temperature.

Types of Research Sources

Research information usually comes from **three main types of sources**:

1. Primary sources
2. Secondary sources
3. Tertiary sources

Each type plays a different role in research.

Understanding these differences helps researchers select **reliable evidence**.

Primary Sources

Primary sources provide original data collected directly by researchers.

They are considered the **strongest type of evidence** because the information is obtained first-hand.

Examples:

- Experiments and measurements
- Surveys and interviews
- Observations
- Prototype testing

Example:
Testing how fast a new processor completes tasks.

Secondary Sources

Secondary sources analyze or interpret primary data.

These sources explain research results and provide expert analysis.

Examples include:

- Journal articles
- Conference papers
- Technical reports
- Review papers

Example:
A research paper analyzing many experiments on machine learning performance.

Tertiary Sources

Tertiary sources summarize knowledge from many studies.

They help researchers understand **basic concepts and background information**.

Examples:

- Encyclopedias
- Dictionaries
- Handbooks

Important note:
They are useful for learning but **should not be the main evidence in research reports.**

Quantitative Data

Quantitative data is numerical data.

It allows researchers to:

- Measure performance
- Compare results
- Test hypotheses
- Identify trends

Example:
A new system reduces processing time by **18%** compared to the previous system.

Numbers make research **objective and measurable**.

Qualitative Data

Qualitative data describes experiences, opinions, or behaviors.

It helps researchers understand **why something happens**, not only what happens.

Examples:

- Interview responses
- User feedback
- Observations

Example:
Users report that an interface is confusing because too many buttons appear on the screen.

Mixed Methods Research

Many research projects combine **quantitative and qualitative data**.

This approach is called **mixed methods research**.

Example:

Quantitative data:

- Task completion time

Qualitative data:

- User opinions about difficulty

Combining both types provides **a deeper understanding of the problem.**

Comparative Analysis

Comparative analysis evaluates multiple options to determine which one performs better.

Researchers define criteria such as:

- performance
- cost
- reliability
- efficiency

Example:
Comparing three cloud storage services based on speed, price, and security.

Cost–Benefit Analysis

Cost–benefit analysis compares the cost of a solution with the benefits it provides.

It helps organizations decide whether a solution is worth implementing.

Example:

Installing solar panels

Cost: $10,000
Savings: $2,000 per year

The company can estimate how long it takes to recover the investment.

Process Analysis

Process analysis studies how a process works step by step.

The goal is to identify:

- inefficiencies
- delays
- errors

Example:
Analyzing the steps in a manufacturing line to find where production slows down.

Improving those steps increases efficiency.

Research Integrity

Research integrity means conducting research honestly and responsibly.

Common integrity violations include:

- Fabricating data
- Ignoring evidence that contradicts conclusions
- Misrepresenting other people's work
- Failing to cite sources

Consequences can include academic penalties or professional damage.

Evaluating Research Sources

Not all sources are reliable.

Researchers must evaluate sources carefully.

Important evaluation criteria include:

- credibility of the author
- reliability of the data
- relevance to the research question
- publication source

Example:
Peer-reviewed journals are usually more reliable than personal blogs.

Confirmation Bias

Confirmation bias occurs when researchers only select information that supports their opinion.

This weakens research because it ignores important evidence.

Example:

If 90% of studies show one conclusion and 10% show the opposite, selecting only the 10% creates a misleading argument.

Researchers must examine **all relevant evidence**.

Defining the Scope of a Project

Research projects often begin with a **very broad topic**.

A clear research scope defines:

- the problem being studied
- the target audience
- available resources
- time limitations

Example:

Broad topic: Artificial Intelligence in Healthcare

Focused research question:
How can AI improve early detection of heart disease?


Thank
you